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SECTION: B

COURSE NAME: FULL STACK WEB DEVELOPMENT

CODE: 23CM4121

TITLE OF THE EXPERIMENT:

To develop an interactive To-Do List using JavaScript, DOM manipulation, and event listeners that allows users to add, complete, and delete tasks dynamically.

DESCRIPTION:-

This experiment demonstrates DOM element selection, dynamic element creation, event listeners, click event handling, and adding and removing elements from a webpage. The user can enter a task, add it to the list, mark it as completed, and delete it.

PROGRAM:-

```
<!-- index.html -->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My To-Do List</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">

  <!-- Page heading -->
  <header>
    <h1> 📌 My To-Do List</h1>
    <p>Organize your tasks and stay productive</p>
  </header>

  <!-- Task input section -->
  <div class="task-input">

    <input type="text"
      id="taskInput"
      placeholder="Enter a task...">

    <button id="addTaskBtn">Add Task</button>

  </div>

  <!-- Tasks are added dynamically here -->
  <div id="taskList"></div>
```

```
<!-- Message shown when the list is empty -->
<p id="emptyMessage">No tasks available.</p>

</div>

<script src="script.js"></script>
</body>
</html>
```

```
/* ===== style.css ===== */

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  min-height: 100vh;
  background: linear-gradient(135deg, #667eea, #764ba2);
  padding: 40px 20px;
}

.container {
  max-width: 700px;
  margin: auto;
}

header {
  text-align: center;
  color: white;
  margin-bottom: 25px;
}

header h1 {
  font-size: 34px;
  margin-bottom: 8px;
}

header p {
  font-size: 15px;
}

.task-input {
  display: flex;
  gap: 10px;
  background: white;
  padding: 20px;
  border-radius: 16px;
  box-shadow: 0 10px 25px rgba(0,0,0,0.2);
  margin-bottom: 20px;
}

#taskInput {
  flex: 1;
  padding: 13px;
  border: 1px solid #ccc;
  border-radius: 8px;
  font-size: 15px;
}

#addTaskBtn {
  padding: 13px 22px;
  border: none;
```

```

    border-radius: 8px;
    background: #667eea;
    color: white;
    font-size: 15px;
    font-weight: bold;
    cursor: pointer;
}

#addTaskBtn:hover {
    background: #5369d8;
    transform: translateY(-2px);
}

.task-item {
    display: flex;
    align-items: center;
    gap: 12px;
    background: white;
    padding: 16px;
    margin-bottom: 12px;
    border-radius: 12px;
    box-shadow: 0 5px 15px rgba(0,0,0,0.15);
}

.task-text {
    flex: 1;
    font-size: 16px;
    font-weight: 500;
}

.complete-btn,
.delete-btn {
    border: none;
    border-radius: 7px;
    padding: 9px 13px;
    color: white;
    font-weight: bold;
    cursor: pointer;
}

.complete-btn {
    background: #28a745;
}

.delete-btn {
    background: #dc3545;
}

.complete-btn:hover {
    background: #218838;
}

.delete-btn:hover {
    background: #c82333;
}

.completed {
    text-decoration: line-through;
    opacity: 0.6;
}

#emptyMessage {
    text-align: center;
    color: white;
    font-size: 16px;
    margin-top: 25px;
}

```

```

/* ===== script.js ===== */

// Select the required HTML elements.
const taskInput = document.getElementById("taskInput");
const addTaskBtn = document.getElementById("addTaskBtn");
const taskList = document.getElementById("taskList");
const emptyMessage = document.getElementById("emptyMessage");

// Update the empty-list message.
function updateEmptyMessage() {
    if (taskList.children.length === 0) {
        emptyMessage.style.display = "block";
    } else {
        emptyMessage.style.display = "none";
    }
}

// Add a new task when the button is clicked.
addTaskBtn.addEventListener("click", function () {

    const taskText = taskInput.value.trim();

    // Prevent adding an empty task.
    if (taskText === "") {
        alert("Please enter a task.");
        return;
    }

    // Create a new task container.
    const taskItem = document.createElement("div");
    taskItem.className = "task-item";

    // Create the task text.
    const task = document.createElement("span");
    task.className = "task-text";
    task.textContent = taskText;

    // Create the Complete button.
    const completeBtn = document.createElement("button");
    completeBtn.className = "complete-btn";
    completeBtn.textContent = "Complete";

    // Mark or unmark the task as completed.
    completeBtn.addEventListener("click", function () {

        task.classList.toggle("completed");

        if (task.classList.contains("completed")) {
            completeBtn.textContent = "Completed";
        } else {
            completeBtn.textContent = "Complete";
        }
    });

    // Create the Delete button.
    const deleteBtn = document.createElement("button");
    deleteBtn.className = "delete-btn";
    deleteBtn.textContent = "Delete";

    // Remove the selected task.
    deleteBtn.addEventListener("click", function () {
        taskItem.remove();
        updateEmptyMessage();
    });
});

```

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});

// Add all elements to the task container.
taskItem.appendChild(task);
taskItem.appendChild(completeBtn);
taskItem.appendChild(deleteBtn);

// Add the task to the webpage.
taskList.appendChild(taskItem);

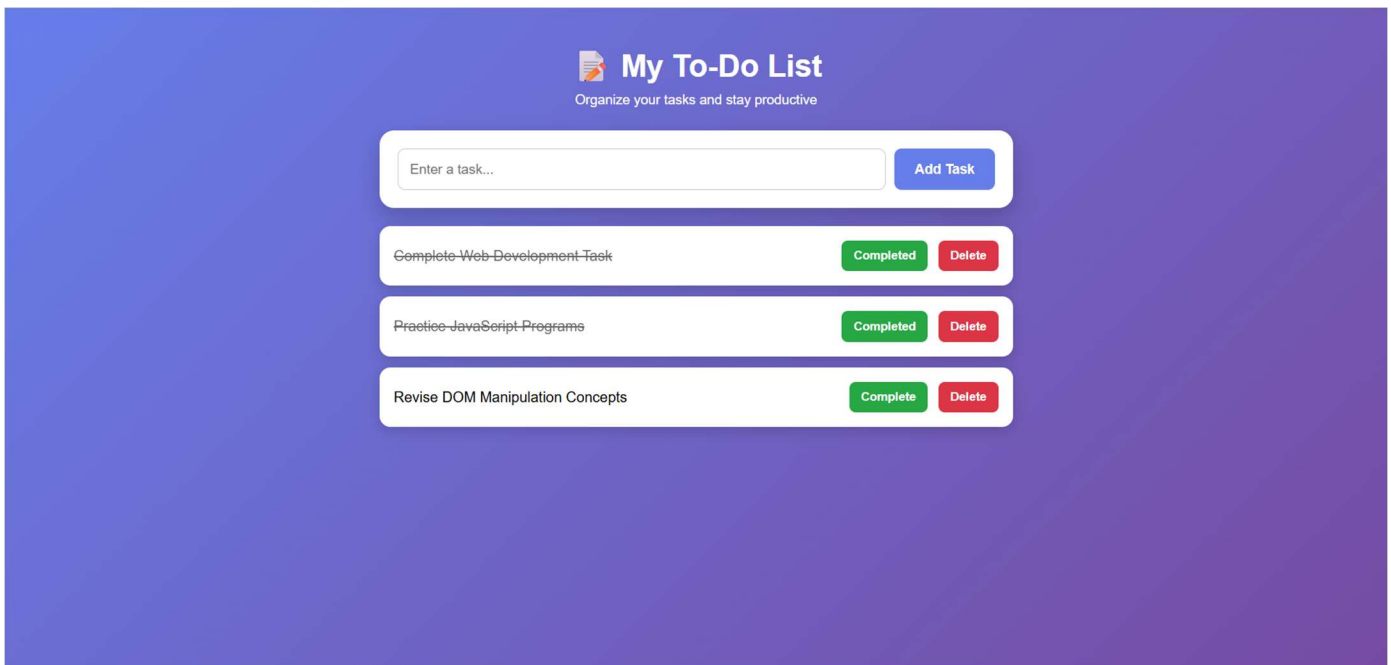
// Clear the input field.
taskInput.value = "";

updateEmptyMessage();
});

// Show the message when the page initially has no tasks.
updateEmptyMessage();

```

OUTPUT:-



THANK YOU